



DR.-ING. JOHANNES ERFURT

AI SYSTEMS ENGINEER | VISION EXPERT

SKILLS

- **Generative AI & Agent Systems**
(LangGraph, Prompt Engineering)
- **Computer Vision & Machine Learning**
(CNNs, ViTs, Fine-Tuning)
- **Software Development & MLOps**
(Python, C++, Docker, CI/CD)
- **Image/Video Processing & Algorithmics**
(Video Codecs, Deep Learning Filters)

PERSONAL PROJECTS

MindGrow Stories | Full-Stack AI Application

- Design and implementation of a cloud-native multi-agent platform for the automated creation of personalized children's books.
- Development of modular AI workflows to orchestrate LLMs, image generation, and VLMs.
- Development of AI-powered methods for automated quality assessment and for ensuring consistent character representation.
- Full-stack development using Next.js, FastAPI, and automated CI/CD-based cloud deployment on Render.
- **Project portfolio:**
<https://johanneserfurt.github.io/projects/mindgrow-stories/>

Wizarding Story Engine | Open-Source Multi-Agent Writing System

- Development of a state-based multi-agent pipeline using LangGraph for planning, generating, evaluating, and iteratively revising longer narrative texts.
- Implementation of structured agent states, specialized critic agents, checkpointing, human-in-the-loop approvals, as well as token and cost monitoring.
- **Project portfolio:**
<https://johanneserfurt.github.io/projects/wizarding-story-engine>

PROFESSIONAL EXPERIENCE

(05/2024 – Present)

Software Developer for AI and Image Processing

 Scansonic MI GmbH, Berlin

Industrial AI & Computer Vision

- **Development of real-time computer vision systems** for automated quality monitoring of industrial laser brazing and welding processes.
- **Training, optimization, and validation of deep learning models** for industrial defect detection and localization, focusing on high generalization and minimization of false-positive rates.
- **Training of cross-site global models** to increase model accuracy and robustness across multiple manufacturing locations.

Pretraining and fine-tuning of modern vision foundation models and implementation of pipelines for image classification and semantic segmentation using CNNs and ViTs.

MLOps & Software Engineering

- **Setup and automation of machine learning pipelines**, including implementation of hyperparameter tuning frameworks and advanced augmentation strategies.
- **Development of a pre-labeling infrastructure** for efficient data preparation and (partial) automation of the annotation process, resulting in an approx. **50% reduction** in manual annotation effort.

Software engineering and deployment infrastructure: Script development for transitioning models into production-ready edge systems (e.g., Triton models).

Technical Leadership & Project Coordination

- **Technical lead for AI software releases:** co-designing technical roadmaps, sprint and release planning, as well as prioritizing and coordinating feature development for on-time software delivery in close alignment with other departments (e.g., Sales, Technical Documentation, Service).
- **Collaboration with international industrial clients** (e.g., Porsche, Tesla, Ford, Stellantis, Daimler Truck) for pilot projects, process optimizations, and on-site/remote AI rollouts.

Professional mentoring, knowledge transfer within the team, and supervision of student theses.

PROFESSIONAL DEVELOPMENT

(11/2023 – 04/2024)

Further Education & Project Phase
(DeepLearning.AI, velpTEC)

Intensive upskilling to prepare for the transition to industrial AI development.

- **Modern AI Stack:** Targeted training in Transformer networks, attention modules, and Large Language Models (LLMs).
- **Computer Vision & Sequence Models:** Architecture and training of deep neural networks (CNNs, RNNs, LSTMs) as well as the application of transfer learning and data augmentation.
- **Technologies:** Deepening applied development skills with PyTorch, TensorFlow, and Scikit-learn.

PROFESSIONAL GAP

(11/2022 – 10/2023)

 World Travel / Parental Leave

LANGUAGES

German – Native
English – Business-fluent
Russian – Very good command

HOBBIES

- Tinkering with custom AI agents 
- Electronic music & dancing 
- Cultural exchange & traveling 
- Table tennis 

PROFESSIONAL EXPERIENCE

(09/2015 – 10/2022)

Research Associate
 Fraunhofer Heinrich Hertz Institute

Research & Algorithm Development (Image and Video Coding)

- **Development and implementation of innovative signal filtering techniques** for image and video compression to increase coding efficiency and subjective quality of H.265/HEVC and H.266/VVC video codecs.
- **Application of mathematical concepts** (wavelets, shearlets, and other sparse transformations) for (in-loop) filtering using MATLAB and C++.
Development of classification and detection algorithms for the use of adaptive Wiener filters as well as for the identification and reduction of coding artifacts.

Machine Learning & Quality Assessment

- **Development of CNN-based machine learning methods** for signal filtering and integration of learned (in-loop and post) filters into the codec simulation software.
- **Optimization of learned filters by combining them with Wiener filters**, achieving an improvement in coding efficiency of more than 4% compared to VVC.
- **Contribution to image quality assessment methods**, particularly the development of the subjective *Weighted Peak Signal-To-Noise Ratio* (WPSNR) measure and subjective-oriented filter optimization.

Software Engineering

- **Development of C++ and Python simulation software** for the automatic evaluation of extensive coding experiments.
- **Maintenance of reproducible development environments** using Git/GitHub for version control, as well as the use of CMake and CI/CD pipelines.

Standardization & Technology Transfer

- **Active participation in VVC standardization (JVET)**, including international presentation and defense of technical contributions at JVET meetings.
- **Co-inventorship on several patent applications** in the addressed research fields.

(08/2014 – 08/2015)

Student Research Assistant
 Fraunhofer Heinrich Hertz Institute

- **Research and development** in the field of video compression with a focus on arithmetic coding algorithms.

(10/2011 – 09/2014)

Student Teaching Assistant
 Technical University of Berlin

- **Conducting tutorials** in mathematics and engineering; explaining complex concepts to students.

SELECTED PUBLICATIONS

- Investigation of Wiener Filter Techniques for In-loop Filtering in Video Coding
J. Erfurt (2023), Doctoral Thesis, Technical University of Berlin
- Shearlet-based Intensity Classification In-loop Filter for Video Coding
J. Erfurt, W. Lim, J. Pfaff, et al. (2023), Signal Processing: Image Communication 115
- Adaptive Loop Filter with a CNN-based Classification
W. Lim, J. Pfaff, B. Stallenberger, J. Erfurt, et al. (2022), IEEE International Conference on Image Processing (ICIP)
- Extended Multiple Feature-based Classifications for Adaptive Loop Filtering
J. Erfurt, W. Lim, et al. (2019), APSIPA Transactions on Signal and Information Processing
- A Study of the Perceptually Weighted Peak Signal-To-Noise Ratio (WPSNR) for Image Processing
J. Erfurt, C. R. Helmrich, et al. (2019), IEEE International Conference on Image Processing (ICIP)

EDUCATION

(09/2019 – 09/2022)

PhD

 Technical University of Berlin

Dissertation: "Filter techniques in video coding"

Final grade: "Very good"

- **Algorithm Development:** Conception and implementation of novel mathematical filter techniques to increase efficiency in video compression.
 - **Method Synergy:** Successful research and combination of classical analytical signal processing with modern machine learning approaches.
 - **Scientific Communication:** Publication of research results in renowned (peer-reviewed) journals and presentation at international conferences.
- Mentoring & Teaching:** Professional guidance of student assistants and intensive supervision of student research and final theses.

(10/2012-03/2015)

Master of Science in Mathematics

 Technische Universität Berlin

Final grade: **1.2**

- Specialization: Signal Processing
- Master's Thesis on "Multidimensional Inpainting" (Grade: **1.0**)

(09/2012 – 01/2013)

Study Abroad

 Tomsk Polytechnic University, Russia

(10/2008 – 09/2012)

Bachelor of Science in Mathematics

 Technische Universität Berlin

Final grade: **1.4**

- Minor: Physics
- Bachelor's Thesis on "Partial Differential Equations"

(09/2010 – 06/2011)

Study Abroad

 Durham University, England

Berlin, 12.08.2026

